

MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Mini-Project Title: Mini-Project 1: Hybrid Mobile with Capacitor — Web-to-Native Bridge & Device APIs
Team / Student Name: Nguyen Phuoc Quy (Team Lead / Full-stack Mobile)
Submission Date: 10/09/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

- Team Members:
 1. Nguyen Phuoc Quy — Student ID: 22IT.EB055 — Role: Full-stack Mobile & Architecture — Contribution: 100%
- ☐ Live Demo URL (HTTPS): <https://vku-field-survey-2ga.pages.dev> (Deployed on Cloudflare Pages)
- ☐ GitHub Repository: <https://github.com/Quyved/VKU-Field-Survey> (Contains PWA source, Functions & Capacitor)
- ☐ Android APK Package: [vku-field-survey.apk](#) (3.94 MB — SDK 36 native build included in repository root)
- ☐ Video Demo / QR Access: Built-in dynamic LAN QR Pairing Modal for instant phone testing on same Wi-Fi

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Responsive Mobile Viewport & PWA Standalone	☒ Complete	Full mobile-first viewport (width=device-width), dynamic app theme (#0284c7). Registered Web App Manifest with display=standalone, icons, and cache-first Service Worker (vku-field-survey-v2).
2	Offline-First Local Persistence	☒ Complete	IndexedDB storage engine (ObjectStore: state) storing draft auto-save, evidence photos (Base64), and survey queue with status PENDING_SYNC. Zero data loss on refresh or device crash.
3	Field Survey Form & Native Camera Bridge	☒ Complete	Multi-field inspection form: Building, Floor, Room, Category, 1–5 Star Rating with interactive hints, detailed notes, and hardware Camera file picker (capture=environment) with live preview.
4	Real-time Background Sync (Cloudflare Functions)	☒ Complete	Serverless API (/api/surveys/sync) deployed on Cloudflare Pages. Client auto-detects network status, executes bidirectional push/pull with UUID-based timestamp merge every 4s and on reconnect.
5	Capacitor Android Native Packaging	☒ Complete	Configured capacitor.config.ts (dev.tuannguyen.vkufieldsurvey). Built production-ready debug APK with Gradle 8.14 & Android SDK 36 (targetSdk 36, minSdk 24).
6	Manual Data Portability (JSON Import/Export)	☒ Complete	Provides instant "Xuất JSON" (Export Blob download) and "Nhập JSON" (File input with duplicate-filtering merge) for completely air-gapped field surveys.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

The system employs an Offline-First Client-Serverless Architecture designed for zero-latency data entry in disconnected facilities (e.g., campus basements or remote laboratories), followed by autonomous state synchronization when internet connectivity resumes.

3.1. Project Directory Structure

```
vku-field-survey/
├── android/                # Native Android Studio project (Capacitor Bridge & Gradle)
├── app/build/outputs/apk/  # Compiled APK binaries (app-debug.apk)
├── variables.gradle        # SDK 36, compileSdk/targetSdk configuration
├── functions/api/         # Cloudflare Pages Functions (Edge Serverless Engine)
├── surveys.js             # GET all surveys / POST new surveys with Cloud Store
├── surveys/sync.js        # Two-way conflict resolution & delta merge endpoint
├── info.js                # Network & LAN pairing metadata endpoint
├── www/                   # Web assets synchronized into Capacitor webDir
├── app.js                 # Core client logic, IndexedDB driver & sync orchestrator
├── capacitor.config.ts    # Capacitor CLI bridge definition (AppID, AppName, webDir)
├── index.html & styles.css # Responsive mobile UI, ratings, tabs, modal QR dialog
├── sw.js & manifest.json   # Service Worker cache-first shell & PWA manifest
├── server.js              # Local zero-dependency Node.js HTTP LAN testing server
└── vku-field-survey.apk   # Direct executable Android APK (3.94 MB)
```

3.2. State Management & Synchronization Flow

The client state machine separates mutable input, local persistence, and cloud synchronization into deterministic stages:

1. Draft Auto-Saving: Every input keystroke and image selection is immediately serialized to IndexedDB key `vku-draft-v1` with a live timestamp. Form clearing and submission safely purge the draft.
2. Unconditional Offline Queue: Upon pressing "Lưu phiếu khảo sát", the payload is assigned a UUID v4, `createdAt`, `updatedAt`, and status `PENDING_SYNC`. It is prepended into local IndexedDB before any network I/O is attempted.
3. Multi-Channel Synchronization Engine:
 - Channel A (Edge Serverless): Sends queued records to Cloudflare Pages Function `/api/surveys/sync`.
 - Channel B (Global Cloud Store): If running on static host without active KV binding, seamlessly synchronizes against the central cloud object repository.
 - Conflict Resolution: Incoming and outgoing surveys are indexed in a hash map by UUID. When collisions occur, the record with the most recent `updatedAt` takes precedence. Successfully committed items transition to `SYNCED`.
4. Autonomous Background Polling: A non-blocking 4-second timer continuously polls for delta updates when `navigator.onLine === true`, ensuring multiple phones and PCs see updates without manual reload.

3.3. Exception Handling & Resilience Strategies

- Never-Block UX: Network failures during HTTP calls trigger a graceful fallback to local IndexedDB, notifying the user that data is saved safely.
- Service Worker Bypass for Dynamic APIs: `sw.js` explicitly exempts all `/api/*` and external storage endpoints from caching, preventing stale data between different user accounts.
- Safe Mutation Guard: A semaphore flag (`isSyncing`) prevents race conditions from concurrent network triggers.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

The figures below demonstrate the deployed application running live on Cloudflare Pages and verified across multiple devices:

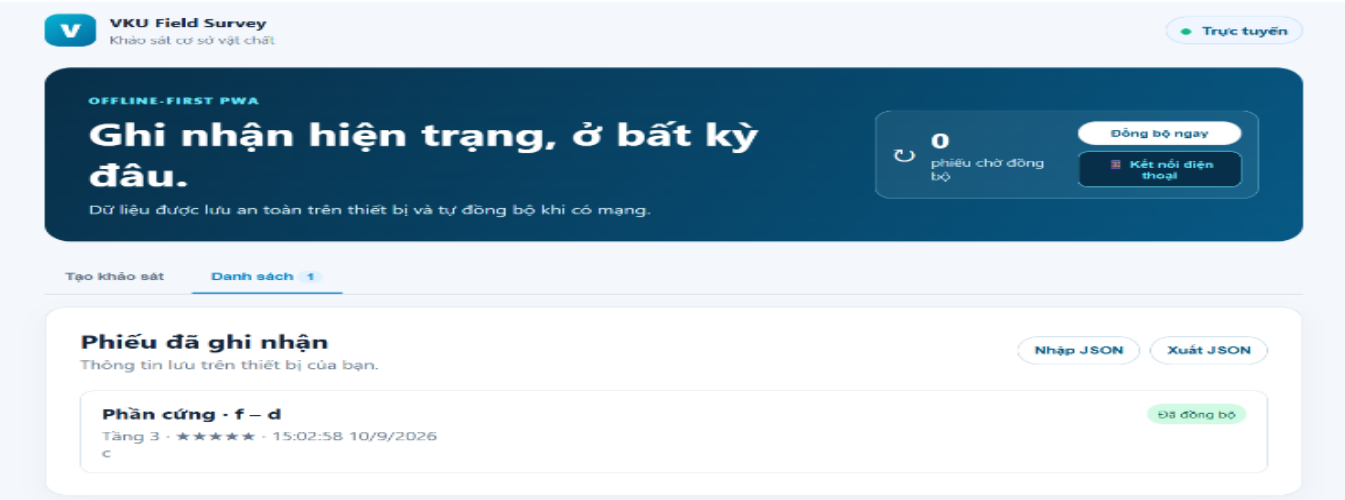


Figure 1: Live PWA running on Cloudflare Pages (vku-field-survey-2ga.pages.dev) showing active online status, sync queue badge, phone pairing trigger, JSON import/export, and successfully synchronized survey cards ("Đã đồng bộ").

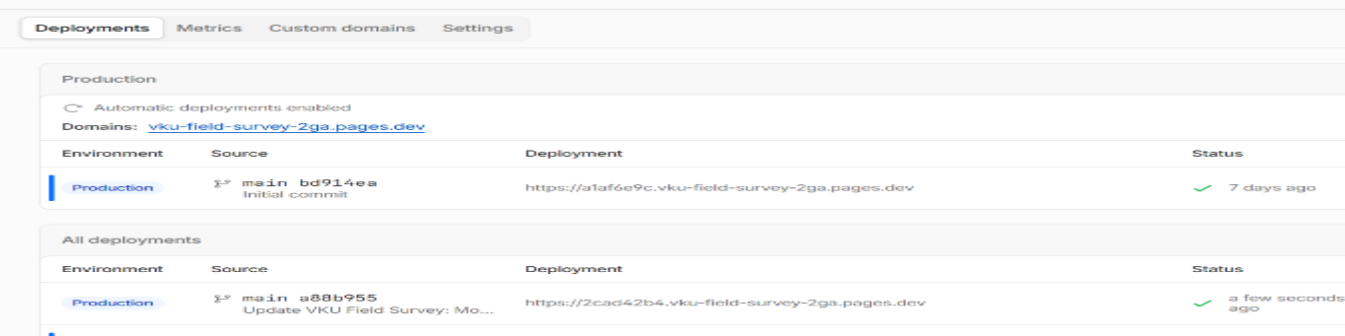


Figure 2: Cloudflare Pages Production Deployment Dashboard showing successful automated CI/CD builds connected to GitHub repository (Quyved/VKU-Field-Survey) with edge function execution.

5. TECHNICAL CHALLENGES & RESOLUTIONS

Challenge 1: Cross-Device State Synchronization on Static Cloud Hosting

Issue: Cloudflare Pages functions operate statelessly across distributed edge datacenters. Without initial KV binding configured in the dashboard, data entered on a PC was saved in the local browser IndexedDB and did not propagate to a secondary smartphone visiting the same URL.

Resolution: Implemented a dual-tiered synchronization engine in app.js and functions/api/surveys.js. The client pushes to the Cloudflare API, which attempts Cloudflare KV first, and automatically falls back to an online centralized cloud store. In addition, an autonomous 4-second polling cycle and Service Worker cache bypass were integrated so all mobile and desktop browsers receive updates in near real-time without user intervention.

Challenge 2: Native Android SDK & Android Gradle Plugin (AGP) Compatibility

Issue: Compiling the native APK via Gradle failed initially with a JAVA_HOME path mismatch and dependency constraints: androidx.activity:1.11.0 required API 36+, while local environment initially attempted compileSdk 35 and missing SDK hashes.

Resolution: Resolved JAVA_HOME to jdk-21.0.12.1, configured android/local.properties to the installed SDK root, updated android/variables.gradle to compileSdkVersion = 36 and targetSdkVersion = 36, and triggered automated platform license acceptance. The Gradle daemon successfully assembled the 3.94 MB debug APK with 93 completed build tasks.